THE SMALLEST EIGENVALUE OF THE NEURAL TANGENT KERNEL OF A DEEP TANH NETWORK IN THE ORDERED PHASE

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Abstract. For a fully connected tanh network of depth L in the infinite-width limit, with weight variance σ_w^2 and bias variance $\sigma_b^2 > 0$ in the ordered phase $\chi_1 < 1$, we determine the smallest eigenvalue of the neural tangent kernel Gram matrix on n distinct inputs of the sphere. It equals $L\chi_1^L\,(q^* + B\,T^*/\chi_1)\,\lambda_{\min}^{\perp}(-A)\,(1 + O(1/L))$, where q^* is the fixed-point preactivation variance, $T^* = q^*/(1-\chi_1)$, $B = \chi'(1)$, and A is the matrix of limiting rescaled correlation gaps. That the non-constant eigenvalues are of order $L\chi_1^L$ is due to Xiao, Pennington and Schoenholz; the contribution here is the constant, its positivity, and the extension to inputs of unequal norm. The factor L comes from a double root in the linear recursion for the kernel gap. We prove that $\lambda_{\min}^{\perp}(-A) > 0$ for any distinct inputs by showing that every Taylor coefficient of the limiting gap profile is strictly positive, so the rate cannot degenerate further.

1. Setting and main result

Let $\phi = \tanh$. A fully connected network of depth L with weights of variance σ_w^2/n_ℓ and biases of variance σ_b^2 has, in the limit of infinite width, preactivations that are centred Gaussian processes with covariance given by the recursion of Poole et al. [5] and Schoenholz et al. [6]

$$\Sigma_{\ell+1}(x,y) = \sigma_w^2\,\mathbb{E}\big[\phi(u)\phi(v)\big] + \sigma_b^2, \qquad (u,v) \sim N\Big(0, \begin{pmatrix} \Sigma_\ell(x,x) & \Sigma_\ell(x,y) \\ \Sigma_\ell(x,y) & \Sigma_\ell(y,y) \end{pmatrix}\Big),$$

and the neural tangent kernel of Jacot, Gabriel and Hongler [3] satisfies

$$\Theta_{\ell+1}(x,y) = \Sigma_{\ell+1}(x,y) + \dot\Sigma_{\ell+1}(x,y)\,\Theta_\ell(x,y), \qquad \dot\Sigma_{\ell+1}(x,y) = \sigma_w^2\,\mathbb{E}\big[\phi'(u)\phi'(v)\big].$$

We assume $\sigma_b^2 > 0$. The diagonal $q_\ell = \Sigma_\ell(x,x)$ obeys $q_{\ell+1} = \sigma_w^2\mathbb{E}[\phi(\sqrt{q_\ell}\,z)^2] + \sigma_b^2$ and converges to a unique fixed point $q^* > 0$. At q^* write $\chi(c) = \sigma_w^2\mathbb{E}[\phi'(u)\phi'(v)]$ for (u,v) of variance q^* and correlation c, and

$$\chi_1 = \chi(1) = \sigma_w^2\,\mathbb{E}\big[\phi'(\sqrt{q^*}z)^2\big], \qquad B = \chi'(1), \qquad T^* = \frac{q^*}{1-\chi_1}.$$

The ordered phase is $\chi_1 < 1$. In it every correlation $c_\ell(x,y) = \Sigma_\ell(x,y)/\sqrt{q_\ell(x)q_\ell(y)}$ tends to 1, and

$$A_{ij} = \lim_{\ell\to\infty} \chi_1^{-\ell}\big(1 - c_\ell(x_i,x_j)\big)$$

exists (Section 2). Let $\mathbf{1}^\perp = \{v \in \mathbb{R}^n : \sum_i v_i = 0\}$ and $\lambda_{\min}^{\perp}(M) = \min\{v^\top M v : v \in \mathbf{1}^\perp,\ |v| = 1\}$.

Theorem 1.1. *Let $x_1,\dots,x_n$ be distinct points with $\Sigma_0(x_i,x_i) = q^*$, and let Θ_L be the $n\times n$ Gram matrix of the neural tangent kernel at depth L. Then, as $L\to\infty$,*

$$\lambda_{\min}(\Theta_L) = \Big(q^* + \frac{B\,T^*}{\chi_1}\Big)\,\lambda_{\min}^{\perp}(-A)\;L\,\chi_1^L\,\big(1 + O(1/L)\big),$$

and $\lambda_{\min}^{\perp}(-A) > 0$. For distinct inputs of arbitrary positive norm the same asymptotic form holds, with A defined through the correlations of the network started at those inputs, whenever $\lambda_{\min}^{\perp}(-A) > 0$.

Remark 1.2. The rate is $L\chi_1^L$ rather than χ_1^L because the gap $\Theta_\ell(x,x) - \Theta_\ell(x,y)$ satisfies a linear recursion whose homogeneous rate and forcing rate are both χ_1; the double root produces the factor L.

For inputs normalized to q^*, Xiao, Pennington and Schoenholz [7] showed that the deviation of Θ_L from its limit, rescaled by $L^{-1}\chi_1^{-L}$, converges and that the eigenvalues of Θ_L other than the largest are $O(L\chi_1^L)$. Theorem 1.1 identifies the limit as $\kappa(-A)$ with $\kappa = q^* + BT^*/\chi_1$ (Section 3), proves that $-A$ is positive definite on $\mathbf{1}^\perp$, so that the order is attained by the smallest eigenvalue and not only bounded above (Section 4), and extends the asymptotics to inputs of unequal norm (Section 5). Hayou, Doucet and Rousseau [2] determined the rates at which the neural tangent kernel approaches its infinite-depth limit for general activations and initializations, and Nguyen, Mondelli and Montúfar [4] gave tight bounds on its smallest eigenvalue for deep ReLU networks.

2. The correlation map and the double root

Fix two inputs x, y with $q_0(x) = q_0(y) = q^*$, so that $q_\ell = q^*$ at every layer, and write $c_\ell = c_\ell(x,y)$ and $D_\ell = 1 - c_\ell$. The correlations obey $c_{\ell+1} = C(c_\ell)$ with

$$C(c) = \frac{1}{q^*}\Big(\sigma_w^2\,\mathbb{E}\big[\phi(u)\phi(v)\big] + \sigma_b^2\Big),$$

(u,v) of variance q^* and correlation c. Differentiating under the Gaussian integral gives $C'(c) = \chi(c)$, and expanding ϕ' in Hermite polynomials shows that $\chi(c) = \sum_{k\ge 0} b_k c^k$ with $b_k \ge 0$. Hence C is convex on $[-1,1]$, $C(1) = 1$ and $C'(1) = \chi_1$.

Lemma 2.1. *In the ordered phase, $D_\ell \to 0$ and the limit $A = \lim_{\ell\to\infty} \chi_1^{-\ell} D_\ell$ exists, with $A > 0$ when $c_0 < 1$ and*

$$D_\ell = A\chi_1^\ell\big(1 + O(\chi_1^\ell)\big).$$

Proof. Convexity and $C'(1) = \chi_1 < 1$ give $C(c) > c$ for $c < 1$, so c_ℓ increases to the only fixed point in $[-1,1]$, which is 1. Taylor's theorem at $c = 1$ gives

$$D_{\ell+1} = \chi_1 D_\ell - \tfrac{B}{2} D_\ell^2 + O(D_\ell^3), \qquad B = \chi'(1) \ge 0,$$

so $D_{\ell+1}/D_\ell = \chi_1(1 + O(D_\ell))$. Since D_ℓ decays geometrically, $\sum_\ell D_\ell < \infty$ and the product $\chi_1^{-\ell} D_\ell = D_0 \prod_{k<\ell} \chi_1^{-1} D_{k+1}/D_k$ converges to a positive limit, the tail of the product being $1 + O(\sum_{k\ge\ell} D_k) = 1 + O(\chi_1^\ell)$. □

The same argument applied to each pair defines the matrix A_{ij} of Theorem 1.1, with $A_{ii} = 0$.

Turn to the kernel. Take $\Theta_0 = \Sigma_0$ and put $T_\ell = \Theta_\ell(x,x)$. On the diagonal $T_{\ell+1} = q^* + \chi_1 T_\ell$, so $T_\ell = T^* + O(\chi_1^\ell)$. Off the diagonal $\Sigma_{\ell+1}(x,y) = q^* c_{\ell+1}$ and $\dot\Sigma_{\ell+1}(x,y) = \chi(c_\ell)$. The gap $E_\ell = T_\ell - \Theta_\ell(x,y)$ therefore satisfies

$$E_{\ell+1} = \chi_1 E_\ell + q^* D_{\ell+1} + \big(\chi_1 - \chi(c_\ell)\big)\,\Theta_\ell(x,y). \tag{1}$$

Proposition 2.2. *With A as in Lemma 2.1,*

$$E_L = A\Big(q^* + \frac{B\,T^*}{\chi_1}\Big) L\chi_1^L\big(1 + O(1/L)\big).$$

Proof. By Lemma 2.1, $q^* D_{\ell+1} = q^* A\chi_1^{\ell+1}(1 + O(\chi_1^\ell))$ and $\chi_1 - \chi(c_\ell) = BD_\ell + O(D_\ell^2)$. A first pass through (1) with the crude bound $|\Theta_\ell(x,y)| \le T_\ell$ gives $E_\ell = O(\ell\chi_1^\ell)$, hence $\Theta_\ell(x,y) = T_\ell - E_\ell = T^* + O(\ell\chi_1^\ell)$. The forcing in (1) is then

$$f_\ell = A\big(q^*\chi_1 + BT^*\big)\chi_1^\ell\big(1 + O(\ell\chi_1^\ell)\big),$$

and solving the recursion,

$$E_L = \chi_1^L E_0 + \sum_{\ell=0}^{L-1} \chi_1^{L-1-\ell} f_\ell = A\big(q^*\chi_1 + BT^*\big)\chi_1^{L-1}\Big(L + O\Big(\sum_{\ell\ge 0}\ell\chi_1^\ell\Big)\Big) + O(\chi_1^L).$$

The sum is finite, which gives the claim. □

For two inputs the Gram matrix has equal diagonal entries, so its smaller eigenvalue is exactly E_L, and Proposition 2.2 is Theorem 1.1 for $n = 2$, where $\lambda_{\min}^{\perp}(-A) = A_{12}$.

3. The n-input Gram matrix

Let $x_1, \dots, x_n$ be distinct with $\Sigma_0(x_i, x_i) = q^*$. Distinct inputs of equal norm have $c_0(x_i, x_j) < 1$, so Lemma 2.1 gives $A_{ij} > 0$ for $i \neq j$, and $A_{ii} = 0$. Put $\kappa = q^* + BT^*/\chi_1$ and let J be the all-ones matrix. Every diagonal entry of Θ_L equals T_L and, by Proposition 2.2 applied to each of the finitely many pairs,

$$\Theta_L = T_L J + \kappa L\chi_1^L(-A) + R_L, \qquad \|R_L\| = O(\chi_1^L). \tag{2}$$

Proposition 3.1. $\lambda_{\min}(\Theta_L) = \kappa\,\lambda_{\min}^{\perp}(-A)\,L\chi_1^L + O(\chi_1^L)$.

Proof. For the upper bound take a unit $u \in \mathbf{1}^\perp$ with $u^\top(-A)u = \lambda_{\min}^{\perp}(-A)$. Since $Ju = 0$, (2) gives $u^\top\Theta_L u = \kappa\lambda_{\min}^{\perp}(-A)L\chi_1^L + O(\chi_1^L)$.

For the lower bound write a unit vector as $w = \alpha n^{-1/2}\mathbf{1} + \beta u$ with $u \in \mathbf{1}^\perp$ a unit vector and $\alpha^2 + \beta^2 = 1$. Then $w^\top Jw = n\alpha^2$, and with $\varepsilon_L = \kappa L\chi_1^L$,

$$w^\top\Theta_L w \;\ge\; nT_L\alpha^2 - 2\|A\|\,\varepsilon_L|\alpha\beta| + \beta^2\varepsilon_L\lambda_{\min}^{\perp}(-A) - O(\chi_1^L).$$

By $2ab \le a^2 + b^2$, $2\|A\|\varepsilon_L|\alpha\beta| \le \frac{1}{2}nT_L\alpha^2 + 2\|A\|^2\varepsilon_L^2\beta^2/(nT_L)$, and the last term is $O(L^2\chi_1^{2L}) = O(\chi_1^L)$. So

$$w^\top\Theta_L w \;\ge\; \tfrac{1}{2}nT_L\alpha^2 + \beta^2\varepsilon_L\lambda_{\min}^{\perp}(-A) - O(\chi_1^L) \;\ge\; \min\big(\tfrac{1}{2}nT_L,\ \varepsilon_L\lambda_{\min}^{\perp}(-A)\big) - O(\chi_1^L),$$

using $\lambda_{\min}^{\perp}(-A) \ge 0$ (shown below). For large L the minimum is $\varepsilon_L\lambda_{\min}^{\perp}(-A)$. □

The constant direction carries the eigenvalue $nT^* + O(L\chi_1^L)$, which stays of order one; every other eigenvalue decays like $L\chi_1^L$, at rates set by the spectrum of $-A$ on $\mathbf{1}^\perp$. Once $\lambda_{\min}^{\perp}(-A) > 0$ is known (Section 4), Proposition 3.1 is Theorem 1.1 for inputs on the fixed-point sphere.

That $-A$ is at least nonnegative on $\mathbf{1}^\perp$ is elementary. For each ℓ the correlations $c_\ell(x_i, x_j)$ form a positive semidefinite matrix K_ℓ with unit diagonal, so $1 - c_\ell(x_i, x_j) = \frac{1}{2}|z_i - z_j|^2$ for some unit vectors z_i, and for $v \in \mathbf{1}^\perp$,

$$v^\top\big(-(J - K_\ell)\big)v = v^\top K_\ell v \ge 0.$$

Dividing by χ_1^ℓ and letting $\ell \to \infty$ gives $v^\top(-A)v \ge 0$. Strictness needs more, because a limit of positive definite matrices can be singular.

4. Strict positivity of $\lambda_{\min}^{\perp}(-A)$

Write the correlation map as a power series. Expanding $\phi(\sqrt{q^*}z) = \sum_k b_k H_k(z)$ in Hermite polynomials, Mehler's formula gives

$$C(c) = \sum_{k\ge 0} a_k c^k, \qquad a_0 = \frac{\sigma_b^2}{q^*}, \qquad a_k = \frac{\sigma_w^2\,k!\,b_k^2}{q^*}\ (k \ge 1),$$

using $b_0 = 0$, since tanh is odd. Thus $a_k \ge 0$, $\sum_k a_k = C(1) = 1$, $a_0 > 0$, and $a_1 > 0$ because tanh is increasing. Infinitely many a_k are positive, since tanh is not a polynomial. In particular $|C(c)| \le C(|c|) \le 1$ on $[-1, 1]$. Put

$$\Phi_\ell(c) = \chi_1^{-\ell}\big(C^{\circ\ell}(c) - 1\big), \qquad c \in [-1, 1],$$

so that $\Phi_\ell(c_0) = -\chi_1^{-\ell} D_\ell$ and, by Lemma 2.1, $\Phi_\ell \to \Phi$ pointwise with $\Phi(c_0(x_i, x_j)) = -A_{ij}$, $\Phi(1) = 0$ and $\Phi < 0$ on $[-1, 1)$.

Lemma 4.1. *There are $p_k \geq 0$ with $\sum_k p_k < \infty$ such that $\Phi(c) = \Phi(0) + \sum_{k\geq 1} p_k c^k$ on $[-1, 1]$, and $\Phi(C(c)) = \chi_1 \Phi(c)$ there.*

Proof. A composition of power series with nonnegative coefficients has nonnegative coefficients, so $\Phi_\ell(c) = \Phi_\ell(0) + \sum_{k\geq 1} p_k^{(\ell)} c^k$ with $p_k^{(\ell)} \geq 0$. Evaluating at $c = 1$, the total mass is $\sum_k p_k^{(\ell)} = -\Phi_\ell(0) \to -\Phi(0)$, so the sequences $p^{(\ell)}$ are bounded in ℓ^1. Any coefficientwise limit point p satisfies $\sum_k p_k c^k = \Phi(c) - \Phi(0)$ for $0 \leq c < 1$ by dominated convergence, which determines p; so $p^{(\ell)} \to p$ coefficientwise. By Abel's theorem and the continuity of Φ at 1 (the convergence in Lemma 2.1 is uniform in c_0, as the error there is bounded by its value at $c_0 = -1$), $\sum_k p_k = -\Phi(0)$, which is the limit of the total masses. By Scheffé's lemma $p^{(\ell)} \to p$ in ℓ^1, so Φ_ℓ converges uniformly on $[-1, 1]$ to $\Phi(0) + \sum_k p_k c^k$, which is therefore Φ. The functional equation follows from $\Phi_{\ell+1} = \chi_1^{-1} \Phi_\ell \circ C$ in the limit. □

Proposition 4.2. *Every p_k, $k \geq 1$, is strictly positive.*

Proof. Both sides of the functional equation are power series with nonnegative coefficients beyond the constant term, converging absolutely on $[0, 1]$, so coefficients may be compared:

$$\chi_1 p_k = \sum_{j\geq 1} p_j \, [c^k]\, C(c)^j \qquad (k \geq 1). \tag{3}$$

The function Φ is not constant, since $\Phi(1) = 0 > \Phi(0)$. It is not a polynomial either: if $p_d > 0$ were the last nonzero coefficient, the right side of (3) would contain $p_d\, C^d$, which has infinitely many nonzero coefficients and cannot be cancelled, while the left side vanishes for $k > d$. Hence $p_j > 0$ for infinitely many j. Given $k \geq 1$, choose such a $j > k$. Selecting k factors $a_1 c$ and $j - k$ factors a_0 in $C(c)^j$ gives $[c^k]\, C^j \geq \binom{j}{k} a_0^{j-k} a_1^k > 0$, so $p_k \geq \chi_1^{-1} p_j \binom{j}{k} a_0^{j-k} a_1^k > 0$. □

Theorem 4.3. *For distinct inputs $x_1, \ldots, x_n$ on the fixed-point sphere, $\lambda_{\min}^{\perp}(-A) > 0$.*

Proof. The matrix $K = (c_0(x_i, x_j))$ is positive semidefinite with unit diagonal, so $c_0(x_i, x_j) = \langle z_i, z_j \rangle$ for unit vectors z_i in some $\mathbb{R}^m$, and they are distinct because $c_0(x_i, x_j) < 1$ for $i \neq j$. By Lemma 4.1, $-A = \Phi(0) J + \sum_{k\geq 1} p_k K^{\circ k}$ with Hadamard powers $K^{\circ k}$. On $\mathbf{1}^\perp$ the first term vanishes. The kernel $\sum_k p_k \langle z, w \rangle^k$ has infinitely many positive even and infinitely many positive odd coefficients, by Proposition 4.2, so it is strictly positive definite on every sphere [8, 1]: the matrix $\sum_k p_k K^{\circ k}$ is positive definite, and its quadratic form is bounded below on the unit sphere of $\mathbf{1}^\perp$. □

5. Inputs off the fixed-point norm

Now let the inputs have arbitrary positive norms. The diagonal recursion $q_{\ell+1} = F(q_\ell)$, $F(q) = \sigma_w^2 \mathbb{E}[\phi(\sqrt{q}\, z)^2] + \sigma_b^2$, converges to q^*, and its rate is $\chi_q = F'(q^*)$.

Lemma 5.1. $0 \leq \chi_q < \chi_1$. *Consequently $|q_\ell(x) - q^*| = O(\chi_q^\ell)$ for every input x.*

Proof. By the heat equation, $\frac{d}{dq} \mathbb{E}[f(\sqrt{q}\, z)] = \frac{1}{2} \mathbb{E}[f''(\sqrt{q}\, z)]$. With $f = \phi^2$ and $u = \sqrt{q^*} z$,

$$\chi_q = \sigma_w^2 \, \mathbb{E}\big[\phi'(u)^2 + \phi(u)\phi''(u)\big] = \chi_1 - 2\sigma_w^2 \, \mathbb{E}\big[\tanh^2(u)\big(1 - \tanh^2(u)\big)\big],$$

using $\tanh'' = -2 \tanh (1 - \tanh^2)$. The last expectation is positive because $q^* > 0$, so $\chi_q < \chi_1$; and $\chi_q \geq 0$ because F is increasing, $\tanh^2$ being increasing in $|u|$. Since F is smooth, $q_{\ell+1} - q^* = \chi_q (q_\ell - q^*)(1 + O(|q_\ell - q^*|))$, and the product of the factors converges. □

Put $\rho = \chi_q$. For a pair x, y the correlations obey $c_{\ell+1} = C_\ell(c_\ell)$, where C_ℓ is the correlation map built from the variances $q_\ell(x), q_\ell(y)$. By Lemma 5.1, C_ℓ and its derivative differ from C and C' by $O(\rho^\ell)$ uniformly on $[-1,1]$; in particular $C_\ell(1) = 1 + O(\rho^\ell)$ and $C_\ell'(1) = \chi_1 + O(\rho^\ell)$. The correlations still tend to 1, because C has 1 as its only fixed point in $[-1,1]$ and attracts every point of it.

Lemma 5.2. *For each pair the limit $A_{xy} = \lim_\ell \chi_1^{-\ell}(1 - c_\ell)$ exists and is finite, and $D_\ell = 1 - c_\ell = A_{xy}\chi_1^\ell + O\big(\chi_1^{2\ell} + \rho^\ell\big)$.*

Proof. Expanding C_ℓ at 1,

$$D_{\ell+1} = \chi_1 D_\ell\big(1 + O(D_\ell)\big) + O(\rho^\ell).$$

Since $D_\ell \to 0$, for any $\lambda \in (\chi_1, 1)$ the factor $\chi_1(1 + O(D_\ell))$ is eventually below λ, so $D_\ell = O(\lambda^\ell)$. The factors $1 + O(D_\ell)$ then have a convergent product, and comparing with the linear recursion gives $D_\ell = O\big(\chi_1^\ell + \sum_{k<\ell} \chi_1^{\ell-k}\rho^k\big) = O(\chi_1^\ell)$, because $\rho < \chi_1$. Dividing by $\chi_1^{\ell+1}$, which is positive since $\phi' > 0$,

$$\chi_1^{-\ell-1} D_{\ell+1} = \chi_1^{-\ell} D_\ell\big(1 + O(\chi_1^\ell)\big) + O\big((\rho/\chi_1)^\ell\big),$$

and both corrections are summable. So $\chi_1^{-\ell} D_\ell$ converges, at the rate of the tails of the two sums. □

The proof of Proposition 2.2 carries over with two changes. The diagonal entries $T_\ell(x) = \Theta_\ell(x,x)$ obey $T_{\ell+1}(x) = q_{\ell+1}(x) + \chi_{1,\ell}(x) T_\ell(x)$ with $\chi_{1,\ell}(x) = \chi_1 + O(\rho^\ell)$, so for two inputs $T_\ell(x) - T_\ell(y)$ satisfies a recursion with rate χ_1 and forcing $O(\rho^\ell)$, whence $T_\ell(x) - T_\ell(y) = O(\chi_1^\ell)$. The gap $E_\ell = T_\ell(x) - \Theta_\ell(x,y)$ satisfies (1) up to an additional forcing $O(\rho^\ell)$, coming from $q_{\ell+1}(x) - \sqrt{q_{\ell+1}(x) q_{\ell+1}(y)}$, from $\chi_{1,\ell}(x) - \chi_1$ and from $\chi_\ell(c_\ell) - \chi(c_\ell)$. Its contribution to E_L is $\sum_{\ell<L} \chi_1^{L-1-\ell} O(\rho^\ell) = O(\chi_1^L)$. Hence

$$E_L = A_{xy}\Big(q^* + \frac{BT^*}{\chi_1}\Big) L \chi_1^L + O(\chi_1^L),$$

and the Gram matrix satisfies (2) with $A = (A_{x_i x_j})$, the differences between the diagonal entries going into R_L. The proof of Proposition 3.1 uses nothing else, so the asymptotic formula of Theorem 1.1 holds whenever $\lambda_{\min}^\perp(-A) > 0$.

The weak inequality survives: the matrix $(c_\ell(x_i, x_j))$ is still positive semidefinite with unit diagonal, and the argument at the end of Section 3 gives $\lambda_{\min}^\perp(-A) \geq 0$. The strict inequality does not follow from Section 4. There $-A_{ij} = \Phi(c_0(x_i, x_j))$ for one function Φ given by a series with positive coefficients, which is what makes the kernel strictly positive definite. Off the sphere each pair is carried by its own sequence of maps C_ℓ, and A is no longer a fixed function of the initial correlations.

6. Numerical checks

All Gaussian expectations below are computed by Gauss–Hermite quadrature, the recursions of Section 1 are iterated exactly in the infinite-width limit, and $\sigma_b^2 = 0.05$ unless stated otherwise. The scripts are in the accompanying repository.

Two inputs. For two inputs on the fixed-point sphere with initial correlation c_0, Proposition 2.2 predicts $E_L/(L\chi_1^L) \to \kappa A$ with a relative error of order $1/L$. Table 1 lists the computed ratio $r_L = E_L/(L\chi_1^L \kappa A)$. In every row $(1 - r_L)L$ is constant to two digits, as an error c/L requires, and $\chi_1^{-L} D_L$ agrees with A to five digits by the last depth shown, in line with the geometric rate of Lemma 2.1.

Eight inputs. Eight random points of the unit sphere in $\mathbb{R}^5$, rescaled to the fixed-point norm, give a 8×8 matrix A whose restriction to $\mathbf{1}^\perp$ is computed from the correlation recursion. The ratio

σ_w^2	c_0	χ_1	κ	A	L	r_L	$(1-r_L)L$
1.5	0.0	0.93864	1.8907	0.40464	160	0.8752	20.0
					200	0.9001	20.0
1.5	0.5	0.93864	1.8907	0.27864	160	0.8901	17.6
					200	0.9121	17.6
1.2	0.0	0.84517	0.4757	0.76500	80	0.9535	3.7
					120	0.9690	3.7
1.2	0.5	0.84517	0.4757	0.42715	80	0.9632	2.9
					120	0.9755	2.9

TABLE 1. The two-input gap against the prediction of Proposition 2.2.

of $\lambda_{\min}(\Theta_L)/(L\chi_1^L)$ to the prediction $\kappa\lambda_{\min}^{\perp}(-A)$ of Theorem 1.1 is

$$0.8171,\ 0.9071,\ 0.9443 \quad (\sigma_w^2 = 1.5,\ L = 60, 120, 200),$$
$$1.0604,\ 1.0311,\ 1.0208 \quad (\sigma_w^2 = 1.2,\ L = 40, 80, 120).$$

Here $|1 - \text{ratio}|\, L$ equals 11.0, 11.1, 11.1 and 2.4, 2.5, 2.5, again an error of order $1/L$. The error changes sign between the two settings, so the approach to the limit is not monotone in general.

The series of Φ. The coefficients p_k of Lemma 4.1 were computed by iterating the composition of truncated power series. At $(\sigma_w^2, \sigma_b^2) = (1.5, 0.3)$, $(1.2, 0.3)$ and $(1.5, 0.05)$ every p_k with $k \le 24$ is positive, the smallest being $1.7 \cdot 10^{-6}$, $1.5 \cdot 10^{-7}$ and $5.7 \cdot 10^{-6}$; the first coefficients are $p_1 = 0.372,\ 0.548,\ 0.200$. This agrees with Proposition 4.2, which needs no numerical input.

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Declaration of competing interest

The author holds equity in Heights Labs and in Convex Nexus Capital and has a management role at Convex Nexus Capital. These entities are included in the funding declaration above.

Declaration of generative AI and AI-assisted technologies

Anthropic Claude AI systems were used to develop the mathematical derivations, write and run the numerical checks, perform internal consistency and literature reviews, and write and prepare this manuscript and its publication files. The author selected the problem and directed the work; the author did not carry out the mathematics or the writing. Responsibility for submission or publication rests with the author. The AI-assisted internal reviews, compilation checks and rendered-page inspections are not external peer review or formal verification.

Data and code availability

The paper is theoretical. The three scripts that produce the numerical checks of Section 6 are in the `code` directory of the repository

`github.com/yspennstate/ntk-smallest-eigenvalue-ordered-phase`.

The paper is part of the Math Brain project, whose public repository is `https://github.com/yspennstate/math-brain-public` [9].

References

[1] D. Chen, V. A. Menegatto and X. Sun, *A necessary and sufficient condition for strictly positive definite functions on spheres*, Proc. Amer. Math. Soc. **131** (2003), 2733–2740.

[2] S. Hayou, A. Doucet and J. Rousseau, *Exact convergence rates of the neural tangent kernel in the large depth limit*, preprint, arXiv:1905.13654.

[3] A. Jacot, F. Gabriel and C. Hongler, *Neural tangent kernel: convergence and generalization in neural networks*, Advances in Neural Information Processing Systems 31 (NeurIPS 2018), 8571–8580, arXiv:1806.07572.

[4] Q. Nguyen, M. Mondelli and G. Montúfar, *Tight bounds on the smallest eigenvalue of the neural tangent kernel for deep ReLU networks*, Proc. 38th Int. Conf. Machine Learning (ICML 2021), arXiv:2012.11654.

[5] B. Poole, S. Lahiri, M. Raghu, J. Sohl-Dickstein and S. Ganguli, *Exponential expressivity in deep neural networks through transient chaos*, Advances in Neural Information Processing Systems 29 (NIPS 2016), arXiv:1606.05340.

[6] S. S. Schoenholz, J. Gilmer, S. Ganguli and J. Sohl-Dickstein, *Deep information propagation*, International Conference on Learning Representations (ICLR 2017), arXiv:1611.01232.

[7] L. Xiao, J. Pennington and S. S. Schoenholz, *Disentangling trainability and generalization in deep neural networks*, Proc. 37th Int. Conf. Machine Learning (ICML 2020), arXiv:1912.13053.

[8] I. J. Schoenberg, *Positive definite functions on spheres*, Duke Math. J. **9** (1942), 96–108.

[9] Y. Shmalo, *Math Brain: public snapshot, code and tests*, 2026, `https://github.com/yspennstate/math-brain-public`.

[10] Y. Shmalo, *Math Brain: shared, auditable memory for multi-agent mathematical research*, October 2026, `https://github.com/yspennstate/math-brain-public/tree/main/papers/math-brain-2026-10-24`.

[11] Y. Shmalo, *Brain Framework*, public software repository, 2026, `https://github.com/yspennstate/brain-framework-public`.